

What is Computer Networking

14 May 2026 15:33

Computer networking refers to connected computing devices (such as laptops, desktops, servers, smartphones, and tablets) and an ever-expanding array of IoT devices (such as cameras, door locks, doorbells, refrigerators, audio/visual systems, thermostats, and various sensors) that communicate with one another.

Main Networking Devices

Switch

- Connects devices inside the same network (PCs, printers, servers).
- Works mainly at **Layer 2**.
- Uses **MAC addresses** to identify devices.
- Helps in internal communication and security.

Access Point (AP)

- A wireless form of switch.
- Connects devices using **Wi-Fi** instead of cables.

Router

- Connects one network to another (for example, home network to Internet).
- Works mainly at **Layer 3**.
- Uses **IP addresses**.
- Chooses the best path for data packets.
- Provides protection from external networks.

MAC Address vs IP Address

MAC Address	IP Address
Physical address of device	Logical network address
Assigned by manufacturer	Assigned by network
Identifies device/NIC	Identifies network connection
Used by switches	Used by routers
Layer 2	Layer 3

A MAC address can be changed, or more accurately, "spoofed" or overridden through software, even though the physical (burned-in) address remains permanently coded into the network hardware. Most modern operating systems allow you to set a "Locally Administered Address" in network adapter settings to hide the real hardware address for privacy or bypass network restrictions.

Important Point

- Modern switches may also support some routing features.

Evolution of Computer Networking

Modern networks are becoming more:

- Programmable
- Automated
- Intelligent
- Secure

A. Software-Defined Networking (SDN)

- Network control is managed by software centrally.
- Makes network:
 - programmable
 - flexible
 - automated
- Helps networks react quickly to changes.

Key Idea

Control plane is separated from hardware devices.

B. Intent-Based Networking (IBN)

- Built on SDN concepts.
- User gives desired outcome (“intent”).
- Network automatically configures itself.

Features

- Automation
- Performance analysis
- Problem detection
- Security integration

C. Virtualized Networks

- One physical network can be divided into multiple logical networks.
- These are called **overlay networks**.

Benefits

- Better security
- QoS customization
- Resource optimization

D. Controller-Based Networking

- Uses centralized controllers.
- Controllers:
 - automate configurations
 - monitor devices
 - improve security
 - simplify management

E. Multidomain Integration

- Different networks/domains communicate together.
- Used in:
 - offices
 - WANs
 - data centers

Purpose

Achieve common business goals across multiple networks.

Types of Computer Networks

Basis	A. LAN	B. MAN	C. WAN
Full-Form	LAN stands for local area network.	MAN stands for metropolitan area network.	WAN stands for wide area network.
Geographic Span	Operates in small areas such as the same building or campus.	Operates in large areas such as a city.	Operates in larger areas such as country or continent.
Ownership	LAN's ownership is private.	MAN's ownership can be private or public.	While WAN also might not be owned by one organization.
Transmission Speed	The transmission speed of a LAN is high.	While the transmission speed of a MAN is average.	Whereas the transmission speed of a WAN is low.
Propagation delay	The propagation delay is short in a LAN.	There is a moderate propagation delay in a MAN.	Whereas, there is a long propagation delay in a WAN.
Congestion	There is less congestion in LAN.	While there is more congestion in MAN.	Whereas there is more congestion than MAN in WAN.
Design & Maintenance	LAN's design and maintenance are easy.	While MAN's design and maintenance are difficult than LAN.	Whereas WAN's design and maintenance are also difficult than LAN as well MAN.
Fault tolerance	There is more fault tolerance in LAN.	While there is less fault tolerance.	In WAN, there is also less fault tolerance.

D. Enterprise Network

Large organizational network.

Requirements

- Highly available
- Scalable
- Robust

Uses

- Campus networks
- Branches
- Data centers

E. Service Provider Network

Operated by telecom/ISP companies.

Provides

- Internet access
- Cellular connectivity
- Leased lines
- Managed services

Basic Terms

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1. Client

- A device or program that requests services/data from a server.
- Examples:
 - Web browser
 - Mobile app

Example

Your browser asking Google for a webpage.

2. Server

- A device or software that provides services/resources to clients.

Examples

- Web server
- Email server
- Database server

3. Peer

- A device in a peer-to-peer network that can both send and receive data.
- No dedicated server required.

Example

Torrent/P2P file sharing.

4. Host

- Any device connected to a network with an IP address.

Examples

- Computer
- Mobile
- Printer
- Server

5. Bandwidth

- Maximum data carrying capacity of a network/channel.
- **Bandwidth**: it refers to the range of frequencies that a transmission medium can support. Higher bandwidth allows higher data transmission rates, assuming other factors such as noise and attenuation remain constant.

Measured In

- bps (bits per second)

Important

Higher bandwidth $\Rightarrow$ more data can be transmitted.

Example

100 Mbps connection.

6. Jitter

- Variation in packet arrival time.

Effects

- Voice/video call disturbance
- Lag in online gaming

Ideal

Lower jitter is better.

7. Packet

- Small unit of data transmitted over a network.

Contains

- Payload (actual data)
- Source & destination info

Important

Large messages are broken into packets.

8. Frame

- Data unit at Data Link Layer (Layer 2).

Contains

- MAC addresses
- Error detection information
- Packet data

Difference

- Packet → Network layer
- Frame → Data link layer

9. Localhost

- Refers to the same computer/device.

IP Address

127.0.0.1

Use

Testing network applications locally.

10. Bit Rate

- Number of bits transmitted per second.

Formula

$$\text{Bit Rate} = \frac{\text{Number of Bits}}{\text{Time}}$$

Unit

- bps, kbps, Mbps, Gbps

11. Noise

- Unwanted electrical/electromagnetic signals affecting communication.

Causes

- Electrical devices
- Radio interference
- Weather

Result

Data corruption.

Interference

- **interference:** It is the disturbance of a signal caused by the addition of unwanted signals from external sources such as electromagnetic radiation or neighboring channels. It degrades signal clarity and may result in data corruption.

12. Attenuation

- Loss of signal strength during transmission.

Causes

- Long distance
- Physical obstacles
- Cable resistance

Solution

- Repeaters
- Amplifiers

13. Distortion

- Change in signal shape/form during transmission.

Causes

- Different frequencies traveling differently
- Channel imperfections

Result

Receiver may interpret wrong data.

14. Transmission Delay

Transmission delay is the time needed to transmit all bits of a packet into the communication channel.

Formula

$$\text{Transmission Delay} = \frac{\text{Packet Size}}{\text{Transmission Rate or Bandwidth (bps)}} = \frac{L}{R}$$

15. Propagation Delay

Propagation delay is the time required for a signal to travel from sender to receiver.

Formula

$$\text{Propagation Delay} = \frac{\text{Distance}}{\text{Propagation Speed}}$$

Signal speed is usually close to:

- speed of light in medium

Connectionless and Connection-Oriented

- **Connectionless Protocol:** Frames are sent from one node to the next without any relationship between the frames; each frame is independent. Connectionless means that there is no connection between frames, it does not imply that there is no physical link between nodes.
- **Connection-Oriented Protocol:** A logical connection should first be established between the two nodes (setup phase). After all frames that are somehow related to each other are transmitted (transfer phase), the logical connection is terminated (teardown phase). The frames are numbered and sent in order. If they are not received in order, the receiver needs to wait until all frames belonging to the same set are received and then deliver them in order to the network layer.

16. Piggybacking

Piggybacking is a technique used in full-duplex communication (both sides can send data simultaneously) to improve efficiency. Instead of sending a separate ACK packet, the receiver attaches (piggybacks) the ACK onto its own outgoing data packet.

Without Piggybacking

Suppose A sends data to B.

A -----> B (Data)

A <----- B (ACK)

Two separate frames/packets are used: Data frame ACK frame

This increases overhead.

With Piggybacking

Suppose B also has data to send to A.

A -----> B (Data)

A <----- B (Data + ACK)

The ACK is included in the header of B's outgoing data frame.

Now only one frame is sent instead of two.

WWW vs Internet

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Definition	A global system of interconnected computer networks that use TCP/IP protocol to link devices worldwide.	A system of interlinked hypertext documents and multimedia content accessible via the Internet..
Scope	All types of digital communication such as gaming, chatting, file sharing, email, etc.	Specifically refers to the collection of web pages and websites accessible via web browsers.
Components	Hardware(servers, routers, devices), protocols(TCP/IP, FTP, SMTP , etc) and various services.	Websites, web pages, web servers and hyperlinks that are used to navigate between them.
Protocols used	TCP/IP, FTS , SMTP, IMAP and many more.	Mainly HTTP/HTTPS
Invention	Originated in the late 1960s and early 1970s as ARPANET	1989 by Tim Berners-Lee
Examples	Email services, VoIP, Online gaming, Cloud storage and more.	Websites like Google, Wikipidea, Youtube, Amazon, Facebook
Usage	Used for all forms of digital communication and data exchange.	Used mainly for accessing information and multimedia using web browsers.

Important Point

The Web is only **one service** on the Internet.

Other Internet services:

- Email
- FTP
- Online gaming
- VoIP
- Cloud storage

Transmission Media

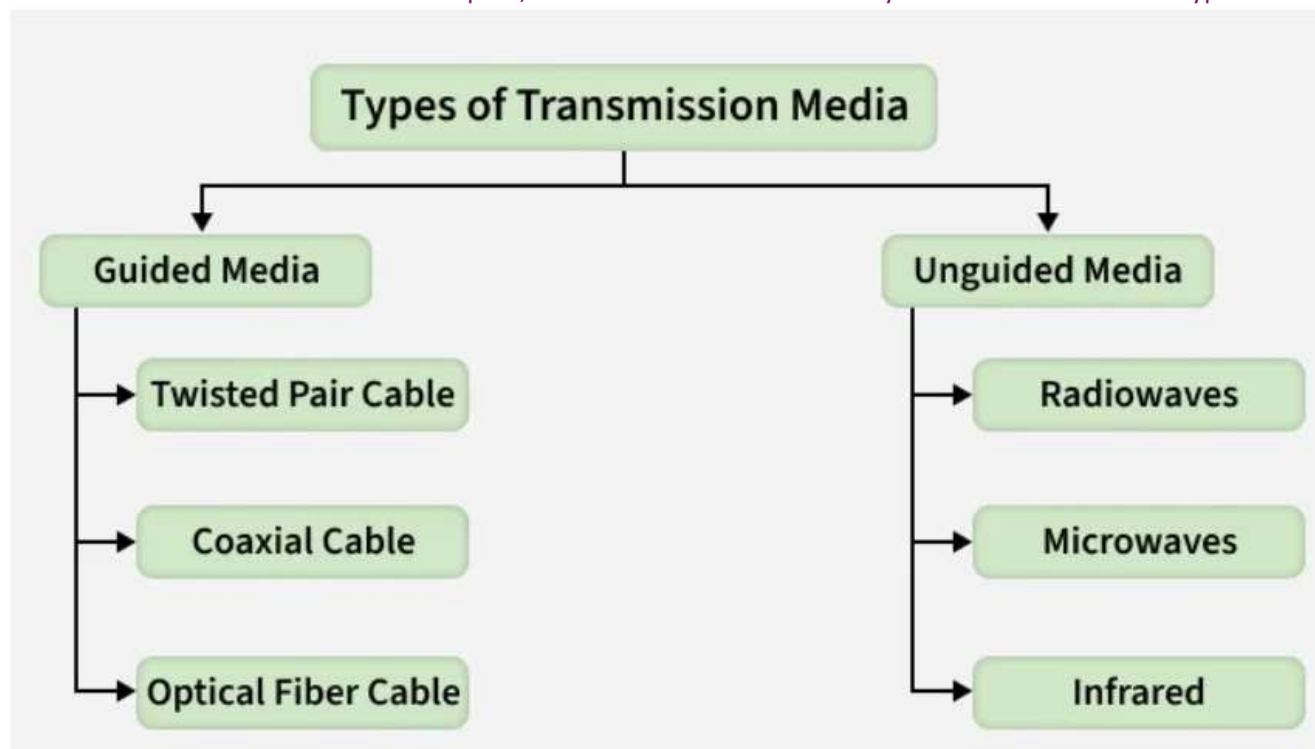
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Transmission media refers to the physical or wireless communication channel used to carry data signals from one device to another within a computer network.

The selection of a transmission medium depends on factors such as:

- Transmission distance
- Data transfer speed (bandwidth)
- Susceptibility to interference and noise
- Cost and installation requirements

Based on the nature of the transmission path, transmission media are broadly classified into two main types:



Guided Media

Guided Media also known as wired or bounded transmission media

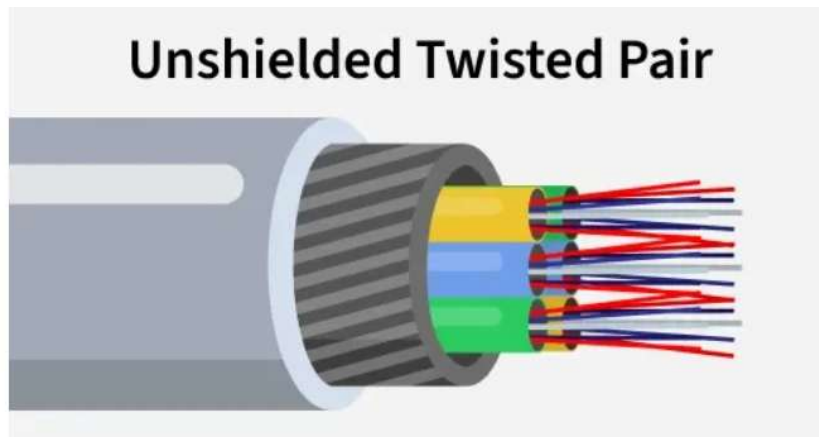
- Uses physical cables to transmit data signals.
- Provides dedicated and well-defined transmission paths.
- Major types of guided media included Twisted Pair Cables, Coaxial Cables and Optical Fiber Cables.
- Offers higher data transmission rates compared to most wireless media.
- Provides better security due to physical connectivity and limited signal leakage.
- Suitable for short to long-distance communication, depending on the cable type used.

1. Twisted Pair Cable

A twisted pair cable consists of two individually insulated copper conductors twisted together in a helical pattern to minimize electromagnetic interference (EMI)

- Twisting of conductors helps reduce interference and signal degradation.
- Multiple pairs can be bundled together within a single cable.
- Widely used due to low cost, flexibility, and ease of installation.
- Commonly used in telephone systems and local area networks (LANs).

(a) Unshielded Twisted Pair (UTP): UTP cable consists of twisted copper wire pairs without any additional metallic shielding. The twisting provides basic protection against interference, making UTP suitable for environments with moderate noise levels.



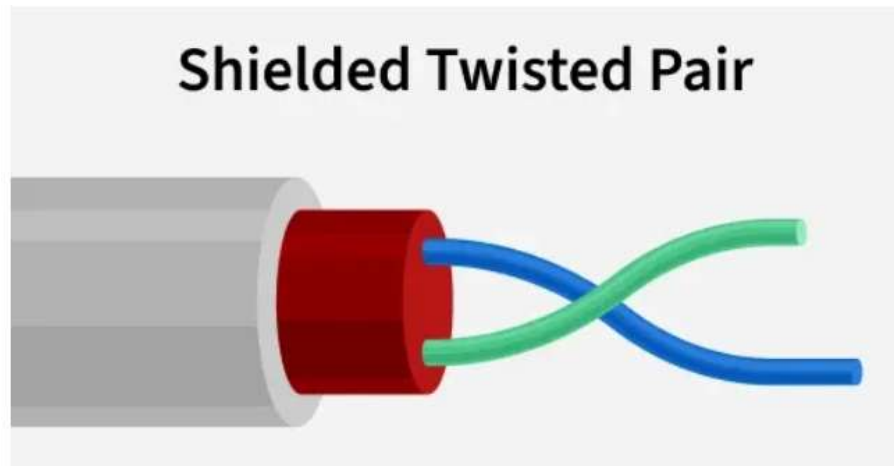
Advantages:

- Least expensive among guided transmission media.
- Easy to install and maintain due to its lightweight and flexible nature.
- Supports high data transmission rates over short distances.
- Widely used in Ethernet-based LANs and telephone networks.

Disadvantages:

- Lower performance and capacity compared to Shielded Twisted Pair (STP).
- Limited transmission distance due to signal attenuation.
- More susceptible to noise and electromagnetic interference.

(b) Shielded Twisted Pair (STP): STP cable consists of twisted pairs of insulated copper conductors enclosed within an additional metallic shielding layer, such as a foil or braided shield. This shielding provides enhanced protection against external electromagnetic interference (EMI) and reduces crosstalk, resulting in more stable and reliable data transmission.



Advantages

- Higher data transmission rates compared to UTP.
- Significantly reduces electromagnetic interference and crosstalk.
- Ensures more reliable and stable communication, especially in noisy environments.
- Suitable for high-speed Ethernet networks and voice and data transmission systems.

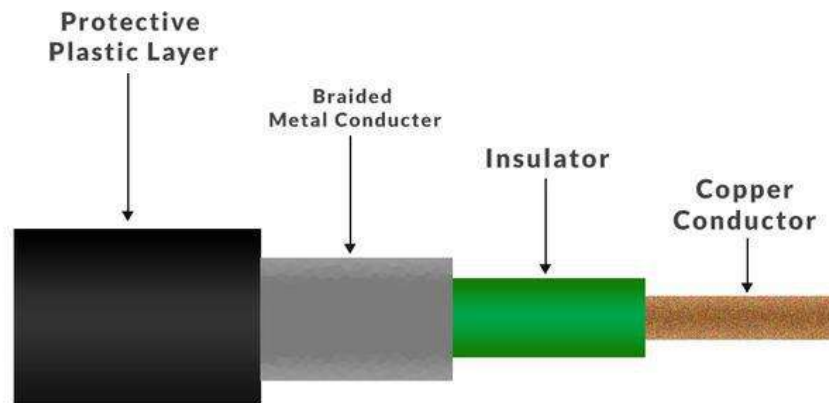
Disadvantages

- Higher cost compared to Unshielded Twisted Pair (UTP).
- More complex installation and termination due to shielding requirements.
- Bulkier and less flexible, making cable management more difficult.

2. Coaxial Cable

Coaxial Cable consists of a central copper conductor surrounded by a dielectric insulating layer, a metallic shielding layer, and an outer protective jacket. This structure allows signals to be transmitted with better protection against noise and interference compared to twisted pair cables.

- Supports data transmission in both baseband mode (single channel) and broadband mode (multiple channels).
- Strong shielding makes it more resistant to noise, crosstalk, and electromagnetic interference (EMI).
- Commonly used in cable television (CATV), broadband networks, and analog television systems.
- Easier to install and maintain compared to optical fiber cables.



Advantages:

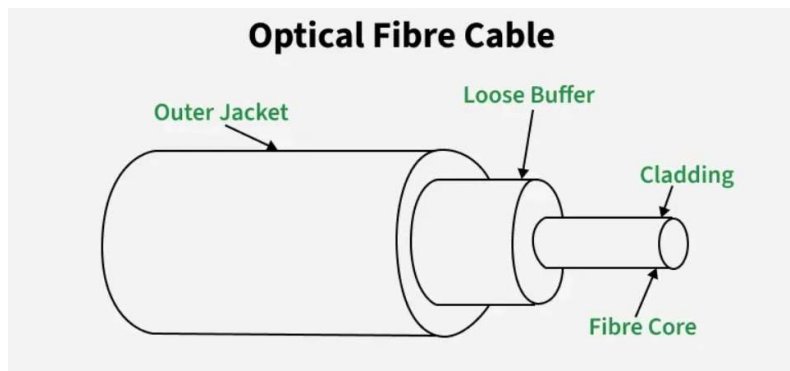
- Supports higher bandwidth than twisted pair cables.
- Offers better signal quality and reliability.
- Allows multiple channels using frequency division multiplexing (FDM).

Disadvantages:

- More expensive than twisted pair cables.
- Requires proper grounding to ensure safety and minimize interference.
- Bulkier and less flexible due to multiple layers.
- More vulnerable to security breaches, as the cable can be physically tapped.

3. Optical Fiber Cable

Optical Fibre Cable is a guided transmission medium that transmits data in the form of light signals through a glass or plastic core using the principle of total internal reflection. The core is surrounded by a cladding layer with a lower refractive index, which confines the light within the core and enables high-speed, long-distance data transmission.



- Supports very large data volumes at extremely high speeds.
- Can operate in unidirectional or bidirectional communication modes.
- WDM (Wavelength Division Multiplexer) allows multiple light signals to be transmitted simultaneously over a single fiber.
- Widely used where high bandwidth, long distance, and low signal loss are required.

Advantages:

- Lightweight and compact, making installation easier over long distances.
- Immune to electromagnetic interference (EMI) and electrical noise.
- Resistant to corrosion and harsh environmental conditions.

Disadvantages:

- Complex installation and maintenance due to precise splicing and handling requirements.
- High initial installation and equipment cost.
- Fragile compared to copper cables and requires careful handling.

Applications:

- **Medical:** Used in endoscopy, imaging systems, and laser-based surgical procedures.
- **Defense and Aerospace:** Supports secure, high-speed, and interference-free communication.
- **Communication:** Widely used in internet backbone networks, undersea cables, and telecommunication systems.
- **Industrial and Automotive:** Used for sensing, lighting, data communication, and safety systems.

Unguided Media

Unguided media, also known as wireless or Unbounded transmission media, uses electromagnetic waves to transmit data without any physical medium. Signals propagate through free space such as air or vacuum. The main types of unguided media are radio waves, microwaves, and infrared waves.

Features:

- Signals propagate through air or free space
- Less secure due to broadcast nature
- Suitable for long-distance communication

1. Radio Waves

Radio Waves are a type of electromagnetic wave that can easily be generated and can propagate through buildings and other obstacles. They do not require line-of-sight between transmitting and receiving antennas, making them highly suitable for broadcast communication and wireless data transmission.

Frequency Range:

3 kHz – 300 GHz

Applications:

AM and FM radio broadcasting, television transmission, cordless phones, and wireless communication.

Types of Radio Waves:

- **Shortwave:** AM radio broadcasting
- **VHF (Very High Frequency):** FM radio and television
- **UHF (Ultra High Frequency):** Television and mobile communication

Components:

- **Transmitter:** Generates and modulates the signal
- **Receiver:** Receives and demodulates the signal

2. Microwaves

Micro Waves use line-of-sight communication, where the transmitting and receiving antennas must be properly aligned. The transmission range depends on the height of the antennas.

Frequency Range:

1 GHz – 300 GHz

Applications:

mobile communication, satellite links, and television distribution.

Advantages:

- Suitable for communication over difficult terrains and oceans
- Supports high data transmission rates

Disadvantages:

- Less secure without proper encryption
- Susceptible to weather conditions such as rain and fog
- Affected by obstacles due to line-of-sight requirement
- High cost of antenna design, installation, and maintenance

3. Infrared

Infrared Waves are used for short-range wireless communication. They cannot penetrate solid obstacles, which limits their range but helps minimize interference between nearby systems.

Frequency Range:

300 GHz - 400 THz

Applications:

TV remote controls, wireless keyboards, mice, and printers.

Causes of Transmission Impairment

Transmission impairment refers to the loss or distortion of signals during data transmission, leading to errors or reduced quality in communication. Common causes include signal distortion, attenuation, and noise all of which can affect the clarity and reliability of transmitted data.

Factors Considered for Designing the Transmission Media

- **Bandwidth**
- **Transmission impairment**
- **Interference**

Applications of Transmission Media in Computer Networks

Transmission Media	Application
Unshielded Twisted Pair (UTP)	Local Area Networks (LAN), telephones
Shielded Twisted Pair (STP)	Industrial networks, environments with high interference
Optical Fiber Cable	Long-distance communication, Internet backbones
Coaxial Cable	Cable TV, broadband Internet, CCTV
Radio	Wireless communication, AM/FM radio, mobile phones
Infrared	Remote controls, short-range communication
Microwave	Satellite communication, radar, long-distance links

Planar Transmission Lines

- Planar transmission lines are specialized guided structures used to carry high-frequency electrical signals over very short distances within electronic circuits and printed circuit boards.
- Although they are not used as network communication media, they apply the same fundamental transmission principles used in guided transmission media, such as controlled signal propagation, impedance matching, and interference reduction.
- Help demonstrate how transmission media principles are applied inside electronic hardware.

Stripline

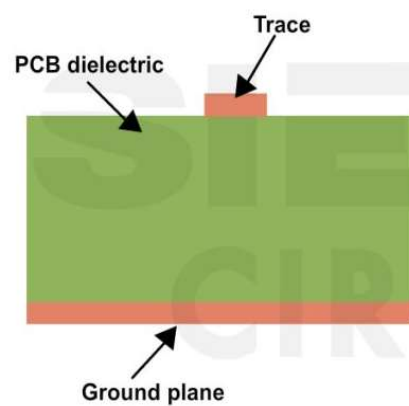
Stripline is a planar guided transmission structure in which a conducting strip is embedded between two parallel ground planes, allowing high-frequency signals to propagate with excellent shielding and minimal electromagnetic interference.

- Fully enclosed structure provides high EMI immunity and stability
- Supports TEM mode propagation (Transverse Electromagnetic (TEM) mode propagation is the transmission of electromagnetic waves where both electric and magnetic fields are purely transverse (perpendicular) to the direction of propagation)
- Used in microwave and high-frequency circuit design
- Suitable for high-performance RF systems

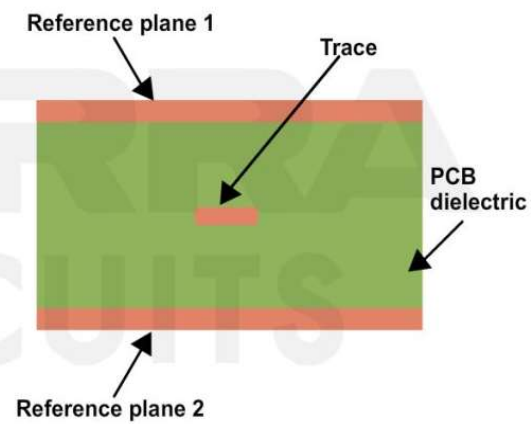
Microstripline

Microstripline is a planar transmission structure consisting of a conducting strip placed on a dielectric substrate with a ground plane on the opposite side, making it widely used for high-frequency signal transmission in printed circuit boards.

- Simple and cost-effective PCB implementation
- Lower shielding than stripline
- Supports quasi-TEM mode propagation
- Common in RF circuits, antennas, and microwave PCBs
- Preferred for compact electronic designs



Microstrip line



Stripline

Network Devices

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1. Hub

works at the **Physical Layer (Layer 1)** of the OSI model.
Its main job is to connect multiple devices physically in a network.

Main Characteristics of Hub

- Connects devices using cables (generally twisted pair cables).
- Sends received data to **all connected devices**.
- Does not check destination address.
- No intelligence or filtering mechanism.
- Works only with electrical signals.
- High chances of collision.
- Usually operates in **half-duplex mode**.



Working of Hub

When a hub receives data:

1. It accepts the electrical signal.
2. Broadcasts the signal to every port.
3. Every connected device receives the data.
4. Only the intended device processes it; others discard it.

Types of Hub

A. Active Hub

Active hubs are smarter than passive hubs.

Functions

- Regenerate signals
- Strengthen signals
- Amplify signals before forwarding

Also Called

- Repeaters

Advantages

- Improves signal quality
- Useful for longer distances

B. Passive Hub

Passive hubs simply act as a connecting point.

Functions

- Only forwards signals
- Does not amplify or regenerate

Characteristics

- No signal improvement
- Simpler and cheaper

Ethernet Hub

An Ethernet hub connects multiple Ethernet devices together. It makes all connected devices behave like a single network segment.

Features of Ethernet Hub

- Uses Ethernet technology
- Works using:
 - CSMA/CD (Carrier Sense Multiple Access with Collision Detection)
- Operates in:
 - Half-duplex mode

Collision in Ethernet Hub

Since all devices share the same communication medium:

- Two devices may send data simultaneously.
- Data collision occurs.

Result

- Lower performance
- Reduced efficiency



2. Switch

A **Switch** is a networking device that connects devices in a LAN more intelligently than a hub.

It mainly works at:

- Data Link Layer (Layer 2)

Some advanced switches also work at Layer 3.

Main Characteristics of Switch

- Connects devices using twisted pair cables.
- Sends data only to intended destination device.
- Uses MAC addresses.
- Reduces collisions.
- Improves network performance.
- Supports full-duplex communication.

Working of Switch

A switch maintains a table of:

- MAC addresses
- Corresponding ports

When data arrives:



A switch maintains a table of:

- MAC addresses
- Corresponding ports

When data arrives:

1. Switch checks destination MAC address.
2. Finds the correct port.
3. Sends data only to that port.



Switching Methods

A. Cut-Through Switching

In this method:

- Switch forwards packet immediately after reading destination address.

Advantages

- Very fast
- Low latency

Disadvantages

- Error checking is limited
- Corrupted packets may be forwarded

B. Store and Forward Switching

In this method:

1. Entire packet is received first.
2. Packet is checked for errors.
3. Then forwarded.

Advantages

- Better error detection
- More reliable

Disadvantages

- Slower than cut-through

C. Fragment-Free Switching

This is a hybrid approach.

Working

- Switch checks initial part of packet.
- Detects collision-related fragments.
- Then forwards packet.

Advantage

- Better than cut-through for collision handling
- Faster than store-and-forward

3. BRIDGES

A **Bridge** at the:

- Data Link Layer (Layer 2) of OSI model.

It connects:

3. BRIDGES

A **Bridge** at the:

- Data Link Layer (Layer 2) of OSI model.

It connects:

- two LANs,
- two network segments,
- or two physical networks using the same protocol.

Main Functions of Bridge

- Connects similar networks.
- Segments large networks into smaller parts.
- Reduces unnecessary traffic.
- Filters frames using MAC addresses.

Working of Bridge

When a bridge receives a frame:

1. It checks the destination MAC address.
2. Determines which segment contains the destination.
3. Forwards frame only if required.
4. Can block unnecessary traffic.

This improves network efficiency.



Learning Bridge

Modern bridges are called **Learning Bridges**.

Function

They automatically learn MAC addresses by observing network traffic.

Advantage

- No need for manual MAC address configuration.

Bridges vs Switches

Switches are considered advanced multi-port bridges.

Why Switches Replaced Bridges

- Faster
- More ports
- Better performance
- More intelligent forwarding

4. ROUTERS

A **Router** works at:

- Network Layer (Layer 3)

It routes packets between different networks.



- Network Layer (Layer 3)

It routes packets between different networks.

Main Functions of Router

- Connects different networks.
- Uses IP addresses.
- Selects best path for packets.
- Controls broadcast traffic.
- Builds large interconnected networks.

Working of Router

When router receives a packet:

1. Reads destination IP address.
2. Checks routing table.
3. Determines best route.
4. Forwards packet to next hop or destination.

Routing Table

A routing table contains:

- destination networks,
- paths,
- next hops.

Routers use routing tables to make forwarding decisions.

Types of Routing

A. Static Routing

Routes are manually configured.

Characteristics

- Simple
- Suitable for small networks
- Manual updates required

Disadvantages

- Time-consuming
- Error-prone
- Difficult in large networks

B. Dynamic Routing

Routes are learned automatically using routing protocols.

Advantages

- Automatic updates
- Better for large networks
- Adapts to topology changes

Common Routing Protocols



- RIP
- OSPF
- EIGRP
- BGP

5. BROUTERS

A **Router** is a combination of:

- Bridge + Router

It performs both functions.

Working of Router

- Acts as a bridge for known/local traffic.
- Acts as a router for other traffic.

Function

- Filters local traffic
- Routes external traffic

Characteristics

- Combines Layer 2 and Layer 3 functionality.
- Rarely used today.
- Functionality mostly integrated into modern routers.

6. GATEWAY

A **Gateway** is a networking device that connects different networks and enables communication between them.

It acts as an entry/exit point between networks.

Main Functions of Gateway

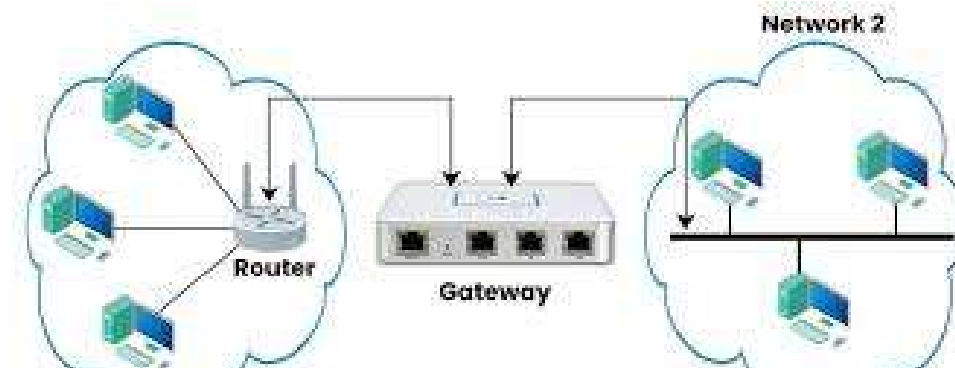
- Connects different networks.
- Translates data between different protocols/formats.
- Allows communication between incompatible systems.
- Passes packets from one network to another.

Working of Gateway

When data arrives:

1. Gateway checks the protocol/format.
2. Converts or translates data if needed.
3. Forwards data to destination network.

Note: Unlike routers, gateways can work at multiple OSI layers.



Note: Unlike routers, gateways can work at multiple OSI layers.

Router as Gateway

A router can also act as a gateway because:

- it forwards packets between networks,
- interprets networking protocols.

However:

- routers mainly perform routing,
- gateways additionally perform translation/conversion.



7. NETWORK CARD (NIC)

A **Network Interface Card (NIC)** is a hardware device that connects a computer to a network.

It is also called:

- LAN card
- Ethernet card
- Network adapter

Functions of NIC

- Establishes physical network connection.
- Converts computer data into electrical signals and vice versa
- Sends/receives data over network.
- Provides MAC address.

Location

NIC is usually:

- installed on motherboard,
- or attached externally.

Types of NIC

A. Wired NIC

- Uses Ethernet cable.

B. Wireless NIC

- Uses Wi-Fi signals.

Modern NICs

Modern NICs:

- are mostly software configurable,
- automatically install drivers,
- support plug-and-play.



System Requirements for NIC

Before installing NIC:

- bus compatibility should be checked.

Common Bus Standard

- PCI (Peripheral Component Interconnect)

Other requirements:

- IRQ settings
- Memory I/O addresses
- Drivers

8. Modem

A **Modem** is a networking device that converts:

- digital signals $\leftrightarrow$ analog signals

so that data can travel through telephone lines.

The word **Modem** comes from:

- **Modulator + Demodulator**

Working of Modem

A. Modulation

Converts:

- Digital signals $\rightarrow$ Analog signals

before transmission through telephone lines.

B. Demodulation

Converts:

- Analog signals $\rightarrow$ Digital signals

when data is received.

Uses of Modem

- Dial-up Internet connection
- Connecting LAN to ISP
- Telephone-line communication

Types of Modems

A. External Modem

Connected externally using:

- USB port
- Serial port

Advantages



- USB port
- Serial port

Advantages

- Easy installation
- Portable
- Automatic resource management
- Useful for laptops and USB users

B. Internal Modem

Installed inside computer as:

- expansion card
- PCMCIA card (for laptops)

Characteristics

- Uses system resources
- Requires manual configuration sometimes

Configuration of Internal Modem

Internal modems require:

- IRQ (Interrupt Request)
- I/O memory addresses

External Modem Configuration

External modems:

- assign resources automatically,
- especially in USB-based systems.

Thus:

- setup becomes simpler.

Factors Affecting Modem Speed

A. UART Speed

UART:

- Universal Asynchronous Receiver/Transmitter

It is a chip responsible for serial communication between:

- computer and modem.

Higher UART speed $\Rightarrow$ better performance.

B. Speed of Modem Hardware

Overall speed also depends on:

- modem's transmission capability.



Message Delivery Options

16 May 2026 15:52

Feature	Unicast	Broadcast	Multicast
Definition	A communication where a message is sent from one sender to one receiver.	A communication where a message is sent from one sender to all receivers.	A communication where a message is sent from one sender to a group of receivers
Transmission	Data is sent to a single recipient	Data is sent to all recipients in a network	Data is sent to a group of recipients
Addressing	Uses a unique destination address	Uses a special broadcast address	Uses a special multicast address
Delivery	Guaranteed delivery	Not all devices may be interested in the data	Not all devices may be interested in the data
Network Traffic	Generates the least amount of network traffic	Generates the most amount of network traffic	Generates moderate network traffic
Security	More secure because data is sent to a specific recipient	Less secure because data is sent to all devices in the network	Moderately secure because data is sent to a specific group of devices
Examples	Email, file transfer	DHCP requests, ARP requests	video streaming, online gaming
Destination	Single receiver	All receivers	Group of receivers
Bandwidth usage	Moderate	High	Moderate
Latency	Low	High	Moderate

Limited Broadcasting

Limited broadcasting is a method where a device sends a message to **all devices in the same local network (LAN)**.

- The packet is **not forwarded by routers**.
- Communication remains inside the local network only.

It is mainly used when a host does not know the destination IP address.

Limited Broadcast Address

The special IP address used for limited broadcasting is:
255.255.255.255

Why Routers Do Not Forward It

Routers block limited broadcasts to:

- avoid unnecessary traffic
- prevent broadcast storms

- keep broadcasts inside one LAN

Directed Broadcasting

Directed broadcasting is a method of sending a packet to **all devices of a specific remote network/subnet**.

Unlike limited broadcast, directed broadcast targets a particular network.

This mode is mainly utilized by television networks for video and audio distribution.

Address Resolution Protocol (ARP) which is used for resolving an IP address into a physical address which is necessary for underlying communication.

Directed Broadcast Address

A directed broadcast address is:

- the **last IP address** of a subnet
- where all host bits are 1

Example

For network: 192.168.1.0/24

Range:

- Network address → 192.168.1.0
- Host addresses → 192.168.1.1 to 192.168.1.254
- Broadcast address → 192.168.1.255

So: 192.168.1.255 is the **directed broadcast address**.

A packet sent to this address reaches all hosts in that subnet.

Network Topology

19 May 2026 16:58

What is Network Topology?

Network topology refers to the arrangement and interconnection of network elements such as computers, links, and nodes. It defines the geometric layout and data flow pattern across the network, directly impacting performance, scalability, and fault tolerance.

There are two main categories:

- Physical Topology: The actual layout of wires, cables, and devices in a computer network.
- Logical Topology: How data flows within the network, regardless of physical design.

Types of Network Topologies

The seven main types of network topology are:

1. Point to Point (P2P) Topology
2. Bus Topology
3. Ring Topology
4. Star Topology
5. Tree Topology
6. Mesh Topology
7. Hybrid Topology

Point to Point (P2P) Topology

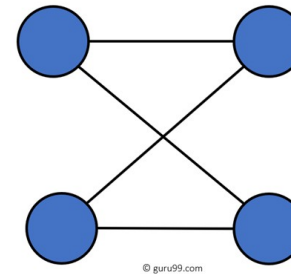
Point-to-Point topology establishes a direct link between two computers or network nodes. The entire bandwidth is available exclusively for the two connected devices.

Advantages of P2P Topology

- Highest bandwidth since the link is not shared with other devices.
- Simple to set up with minimal technical expertise.
- Easy to troubleshoot with only two devices and one connection.

Disadvantages of P2P Topology

- Only practical for connecting two devices, unsuitable for larger networks.



Bus Topology

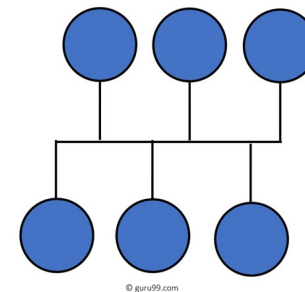
In Bus topology, all devices connect to a single central cable known as the bus or backbone. Data sent by any device travels in both directions until it reaches its destination.

Advantages of Bus Topology

- Cost-effective for small networks, requiring less cable than other topologies.
- Easy to implement and extend by adding devices to the backbone.
- Does not require specialized hardware such as hubs or switches.

Disadvantages of Bus Topology

- Backbone cable failure disrupts the entire network.
- Performance degrades as more devices are added due to data collisions.
- Troubleshooting is difficult since all devices share one line.
- Limited cable length restricts network size.



Ring Topology

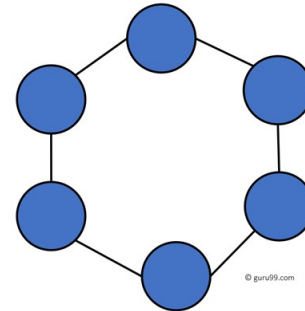
In Ring topology, every device has exactly two neighbors. Data travels in one direction around the ring, passing through each node until it reaches its destination. A token-passing mechanism regulates transmission.

Advantages of Ring Topology

- Data flows in one direction, reducing packet collisions.
- Equal access ensures no single node monopolizes the network.
- Token-passing protocol provides orderly data transfer.

Disadvantages of Ring Topology

- Adding or removing devices requires shutting down the network temporarily.
- Slower than star topology because data passes through each intermediate node.
- A single break halts all communication unless a dual-ring is used.



Star Topology

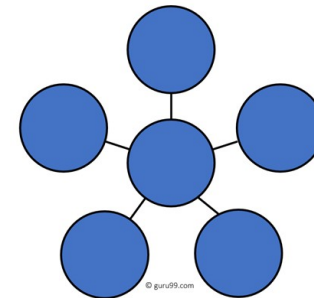
In Star topology, all devices connect to a central hub or switch. Each node has a dedicated connection to the hub, and all data passes through it to reach other devices.

Advantages of Star Topology

- Easy to troubleshoot with dedicated connections per device.
- Failure of one node does not affect the rest of the network.
- Simple to add or remove devices without disruption.

Disadvantages of Star Topology

- If the central hub fails, the entire network goes down.
- Performance depends on the capacity of the central hub.



Mesh Topology

In Mesh topology, each computer connects to every other computer, creating multiple data pathways. If one link fails, data reroutes through an alternative path.

Types of Mesh Topology

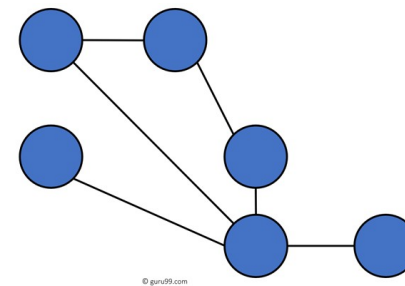
- **Partial Mesh:** Some devices connect to all others, while certain devices connect only to those they exchange the most data with.
- **Full Mesh:** Every device is connected directly to every other device. This provides maximum redundancy but at a higher cost.

Advantages of Mesh Topology

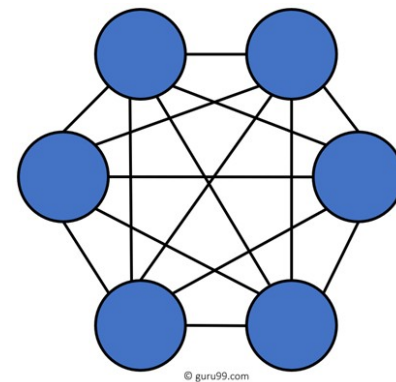
- Maximum redundancy: if one path fails, data reroutes through another.
- Highly reliable for mission-critical applications like banking and healthcare.

Disadvantages of Mesh Topology

- Requires many cables and I/O ports, making it expensive.
- Installation and configuration are complex.
- Full mesh is impractical for large networks due to exponential connection growth.



Partial Mesh



Tree Topology

Tree Topology

Tree topology organizes devices in a hierarchical structure combining star and bus topologies, suitable for large, segmented networks.

Advantages of Tree Topology

- Easy to manage and segment large networks hierarchically.
- Error detection is straightforward at each level.
- Branches can be managed independently.

Disadvantages of Tree Topology

- Heavily cabled and more expensive than bus topology.
- Root node or backbone failure affects the entire segment.
- Difficult to maintain as the network grows in complexity.

Hybrid Topology

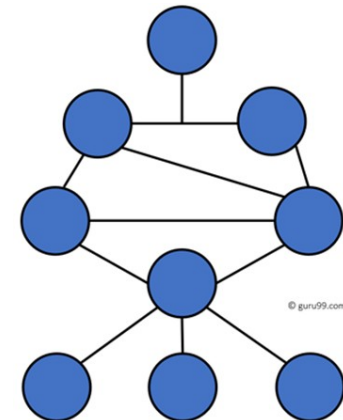
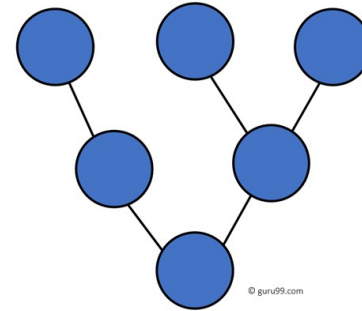
Hybrid topology combines two or more topology types into a single network, allowing designers to leverage the strengths of each type.

Advantages of Hybrid Topology

- Flexible and scalable without redesigning entire infrastructure.

Disadvantages of Hybrid Topology

- Complex design requiring experienced network engineers.
- More expensive due to diverse hardware requirements.



Full Mesh

Transmission Modes

16 May 2026 16:14

Transmission Modes describe how data is exchanged between two devices over a communication channel. They specify whether information moves in a single direction or in both directions, and whether devices can transmit at the same time.

- Defines the pattern of communication
- Determines sender and receiver roles
- Influences speed and performance

Simplex Mode

Simplex Mode is a transmission mode in which communication occurs in only one direction, from a sender to a receiver, without any possibility of reverse data flow.

- One device acts only as transmitter
- The other device acts only as receiver
- No feedback or acknowledgment is supported

Example: keyboard and traditional monitors. The keyboard can only introduce input, the monitor can only give the output.

Advantages

- Simple and easy to implement
- Lower installation and maintenance cost
- Utilizes the entire bandwidth for sending
- Suitable for broadcast-type applications

Disadvantages

- Not suitable for interactive systems
- Cannot confirm successful delivery

Half-Duplex Mode

Half-Duplex Mode is a transmission mode in which communication can occur in both directions, but only one device can transmit at a time. The devices take turns sending and receiving data over the same channel.

- Uses a single shared communication channel

Example: walkie-talkie in which message is sent one at a time and messages are sent in both directions.

Advantages

- Efficient use of a single channel
- Suitable for controlled communication environments

Disadvantages

- Cannot send and receive simultaneously
- Messages delay due to turn-based transmission

- Possible delay due to turn-based transmission
- Performance decreases with heavy traffic
- Risk of collision if control mechanisms fail

Full-Duplex Mode

Full-Duplex Mode is a transmission mode in which communication takes place in both directions at the same time. Both connected devices can transmit and receive data simultaneously without waiting for each other.

- Uses separate channels or divided bandwidth

Example: Telephone network in which there is communication between two persons by a telephone line, through which both can talk and listen at the same time.

Advantages

- Faster data transfer rate
- No waiting time between transmissions
- Improved communication efficiency
- Suitable for interactive applications

Disadvantages

- Higher installation cost
- Requires more complex hardware
- Greater bandwidth requirement
- Increased system configuration complexity

Other Terms

16 May 2026 17:03

NETWORK PROTOCOLS

Network protocols are a set of:

- rules,
- conventions,
- instructions

that govern communication between network devices.

Purpose of Protocols

Protocols define:

- how data is transmitted,
- how devices communicate,
- error handling,
- synchronization,
- formatting rules.

Important Point

For communication:

- devices must use compatible protocols.

Examples of Network Protocols

Protocol	Purpose
HTTP	Web communication
HTTPS	Secure web communication
TCP/IP	Internet communication
FTP	File transfer
SMTP	Email sending
DNS	Domain name resolution

Multiple Protocol Support

A computer can support:

- multiple protocols simultaneously.

Example:

- HTTP for web browsing

- FTP for file transfer
- SMTP for email

TCP/IP

TCP/IP is the foundation of the Internet.

TCP

- Reliable data transfer

IP

- Addressing and routing

ISDN (Integrated Services Digital Network)

ISDN is a WAN communication technology used for transmitting:

- voice,
- audio,
- graphics,
- data

over telephone networks digitally.

It was introduced as a faster alternative to traditional dial-up connections.

Main Features of ISDN

- Digital communication technology
- Higher speed than traditional modem connections
- Faster connection establishment
- Supports multiple communication channels on one line

Uses of ISDN

- Internet access
- Voice communication
- Video conferencing
- Data transfer

Working of ISDN

ISDN converts traditional telephone lines into digital communication lines.

It allows:

- simultaneous transmission of voice and data.

Advantages of ISDN

Advantage	Description
Faster than dial-up	Better speed
Digital transmission	Better quality
Quick connection setup	Faster line pickup
Multiple channels	Supports simultaneous communication

Disadvantages of ISDN

- Expensive compared to modern broadband
- Availability depends on service provider
- Slower than modern technologies like fiber/DSL

Present Status

ISDN is mostly outdated today because:

- broadband,
- fiber optics,
- DSL,
- mobile internet

are faster and cheaper.

Difference between Internet, Intranet and Extranet

Point of Difference	Internet	Intranet	Extranet
Accessibility	Public – open to all	Private – only employees	Private – employees + authorized external users
Coverage	Worldwide	Within an organization	Organization + selected partners/clients
Purpose	Global information sharing	Internal communication & collaboration	Secure collaboration with external stakeholders
Security	User - dependent; less controlled	Protected by organizational firewalls	Firewalls + encryption + controlled login
Ownership	No single owner	Single organization	Single or multiple organizations
Users	General public	Employees of the organization	Employees + suppliers/partners/clients
Maintenance	Maintained by ISPs and global organizations	Maintained by the organization's IT team	Maintained jointly by the organization and external partners
Example	Public web browsing	Wipro's internal HR portal	Dell–Intel supplier network

